Integration: Advanced Methods (Trigonometric)

Questions 1–31

Question 1.

a) Use the substitution $x = \tan u$ to find $\displaystyle\int \frac{x^2}{1+x^2}\,dx$.

b) By writing $\dfrac{x^2}{1+x^2}$ in the form $A + \dfrac{B}{1+x^2}$, where A and B are integers, confirm your result from part a).

Question 2. Use the substitution $x = \tan u$, or otherwise, to find each of these.

a) $\displaystyle\int \frac{2x^2-1}{1+x^2}\,dx$

b) $\displaystyle\int \frac{5x^2+9}{1+x^2}\,dx$

Question 3. Find $\int \frac{1}{\sqrt{x(1+x)}}\,dx$ by:

a) using the substitution $\sqrt{x} = \tan u$

b) using the substitution $\sqrt{x} = u$

Question 4. Find $\int \frac{1}{(1+x^2)^{3/2}}\,dx$.

Question 5. Use the substitution $u = \sqrt{x^2 - 1}$ to find $\displaystyle\int \frac{1}{x\sqrt{x^2-1}}\,dx$.

Question 6.

a) Write $\sin^2\theta$ in terms of $\cos 2\theta$.

b) Use the substitution $x = \sin u$ to find $\displaystyle\int \frac{x^2}{\sqrt{1-x^2}}\,dx$.

Question 7. Use a suitable substitution to find each of these.

a) $\int \frac{x^2}{\sqrt{1-4x^2}}\,dx$

b) $\int \frac{x^2}{\sqrt{4-x^2}}\,dx$

c) $\int \frac{x^2}{\sqrt{4-2x^2}}\,dx$

Question 8. Use a suitable substitution to find each of these.

a) $\int \frac{2x^2+3}{\sqrt{1-x^2}}\,dx$

b) $\int \frac{3-x^2}{\sqrt{1-9x^2}}\,dx$

c) $\int \frac{3x^2+1}{\sqrt{1-3x^2}}\,dx$

Question 9. Find the exact value of $\displaystyle\int_{-1/4}^{1/4} \frac{8+8x^2}{\sqrt{1-4x^2}}\,dx$.

Question 10.

a) Show that $\dfrac{d}{dx}\left(3x\sqrt{1-9x^2} - \arccos(3x)\right) = k\sqrt{1-9x^2}$, where k is a positive integer to be found.

b) Hence find the exact value of $\displaystyle\int_{1/6}^{\sqrt{3}/6} \sqrt{1-9x^2}\,dx$.

Question 11.

a) Show that

$$\frac{d}{dx}\left(2(x-5)\sqrt{11+10x-x^2}+72\arcsin\left(\frac{x-5}{6}\right)\right)=k\sqrt{11+10x-x^2},$$

where k is a positive integer to be found.

b) Hence find the exact value of $\displaystyle\int_5^8 \sqrt{11+10x-x^2}\,dx$.

Question 12.

a) Write $19+18x-x^2$ in the form $p-(x+q)^2$, where p and q are integers.

b) Using your answer to part a), use the substitution $\dfrac{x+q}{\sqrt{p}}=\sin u$ to show that

$$\int \sqrt{19+18x-x^2}\,dx=\frac{1}{2}(x-9)\sqrt{19+18x-x^2}+50\arcsin\left(\frac{x-9}{10}\right)+c.$$

Question 13. Use a suitable substitution to find $\int \sqrt{3+2x-x^2}\,dx$.

Question 14.

a) Show that $\sin 3A \equiv 3\sin A - 4\sin^3 A$.

b) Show that $\cos 3A \equiv 4\cos^3 A - 3\cos A$.

c) Use a suitable substitution to find $\int \dfrac{6x^3-4x+1}{\sqrt{1-x^2}}\,dx$.

Question 15.

a) Find $\dfrac{d}{dx}\left(\arctan\sqrt{\dfrac{x}{3}}\right)$.

b) Hence show that $\displaystyle\int_1^3 \frac{\sqrt{x}}{3x+x^2}\,dx = \frac{\pi}{6\sqrt{3}}$.

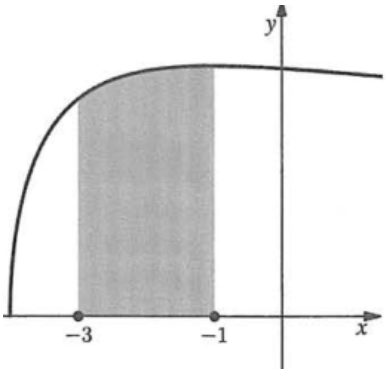
Question 16. Use the substitution $u^2 = x-3$, where $u > 0$, to find $\displaystyle\int \frac{\sqrt{x-3}}{x+6}\,dx$.

Question 17. Use the substitution $u^2 = x + 1$, where $u > 0$, to find $\displaystyle\int \frac{x}{\sqrt{x+1}(x+10)}\,dx$.

Question 18. The curve shown in the diagram below has equation

$$y = \frac{\sqrt{x+4}}{x+7}, \qquad x \geq -4.$$

The shaded region is the region bounded between the curve and the x-axis, and the lines $x = -3$ and $x = -1$. Show that the exact area of the shaded region is $\left(2\sqrt{3} - 2 - \frac{\pi}{2\sqrt{3}}\right)$ units^2.



Question 19. Use integration by parts to find:

a) $\int \arcsin x \, dx$

b) $\int \arccos x \, dx$

c) $\int \arctan x \, dx$

Question 20. Show that $\int_0^1 8 \arctan x \, dx = 2\pi - \ln p$, where p is a positive integer to be found.

Question 21. Find:

a) $\int \arcsin 2x \, dx$

b) $\int \arccos 3x \, dx$

c) $\int \arctan 4x \, dx$

Question 22. Find:

a) $\int 8x \arcsin(x^2) \, dx$

b) $\int e^{2x} \arccos(e^{2x}) \, dx$

c) $\int \sin x \arctan(\cos x) \, dx$

d) $\int \sec^2 x \arcsin(\tan x) \, dx$

e) $\int 3^x \arctan(3^x) \, dx$

Question 23. Find:

a) $\int x \arcsin x \, dx$

b) $\int x^2 \arccos x \, dx$

c) $\int x^3 \arctan x \, dx$

Question 24.

a) Write $\dfrac{10x^2}{1+x^2}$ in the form $A + \dfrac{B}{1+x^2}$.

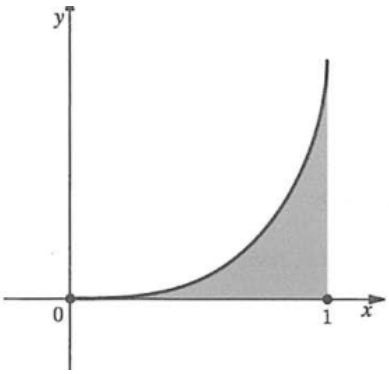
b) Find $\int 20x \arctan x \, dx$.

Question 25.

a) Write $\dfrac{15x^2}{4+x^2}$ in the form $A + \dfrac{B}{4+x^2}$.

b) Find $\displaystyle\int 30x \arctan\left(\frac{x}{2}\right)\, dx$.

Question 26. The curve shown in the diagram below has equation $y = 6x\arcsin(x^2)$, $0 \leq x \leq 1$. Find the exact area of the shaded region.



Question 27. Find:

a) $\displaystyle\int \frac{\arcsin x}{\sqrt{1-x^2}}\,dx$

b) $\displaystyle\int \frac{\arccos x}{\sqrt{1-x^2}}\,dx$

c) $\displaystyle\int \frac{\arctan x}{1+x^2}\,dx$

Question 28. Show that $\displaystyle\int_{1/2}^{1/\sqrt{2}} \frac{(\arcsin x)^2}{\sqrt{1-x^2}}\,dx = k\pi^3$, where k is a rational number to be found.

Question 29. Find $\int \arctan x \ln(1+x^2)\,dx$.

Question 30. Find:

a) $\int \operatorname{arccosec} x\,dx, \quad x > 0$

b) $\int \operatorname{arcsec} x\,dx, \quad x > 0$

c) $\int \operatorname{arccot} x\,dx$

Question 31. The curve shown in the diagram below has equation $y = \operatorname{arccosec} 2x$ for $x > \frac{1}{2}$. The shaded region is the region bounded between the curve and the x-axis, and the lines $x = \frac{1}{\sqrt{2}}$ and $x = 1$. Show that the exact area of the shaded region is

$$\left(\frac{1}{2}\ln\left(\frac{2+\sqrt{3}}{1+\sqrt{2}}\right) + \frac{\pi}{6} - \frac{\pi}{4\sqrt{2}}\right) \text{ units}^2.$$

